

CAREER: Decoupling Graph Mining from GPU Execution through Self-Adaptive Primitives

Overview

Graph mining, spanning diverse tasks such as clique discovery, motif mining, subgraph matching, and community detection, powers fraud detection, drug discovery, cybersecurity, and modern AI data pipelines. Driven by extreme algorithmic complexity and ever-growing data volumes, modern graph mining increasingly depends on high-performance computing. With their massive parallelism and high memory bandwidth, GPUs have emerged as the leading platform for accelerating graph mining. Unlike CPU-based parallel graph mining, where recursive search and task scheduling are largely mediated by mature operating-system, compiler, and runtime abstractions, CUDA-based GPU graph mining exposes the programmer directly to the accelerator execution model. Algorithm designers must manually choose warp/block mappings, memory layouts, synchronization points, buffer management strategies, and launch configurations. This tight coupling between mining logic and GPU execution mechanics makes current systems difficult to reproduce, tune, and port across GPU generations and even different datasets. Also, scaling beyond a single GPU typically requires a near-complete rewrite. To bridge this gap, this project develops a set of GPU-native graph-mining primitives that decouple mining logic from GPU execution, so that a single mining program auto-tunes across GPU generations and scales to multiple GPUs without rewriting.

Intellectual Merit

The project advances GPU-accelerated graph-mining with three thrusts. First, it develops an abstraction that separates mining logic from GPU execution mechanics, and shows that a set of primitives, including vertex ordering, neighbor intersection, candidate expansion, pruning, verification, and enumeration, suffices to express diverse mining algorithms such as clique enumeration, motif counting, k -truss decomposition, subgraph matching, and quasi-clique mining, with explicit warp-, block-, and device-level mappings. Second, it casts performance portability as a learnable search over hardware features (number of compute units, memory hierarchy, memory bandwidth, tensor cores) and graph features (degree skew, density, locality), with auto-tuning algorithms that exploit mining-specific structure such as search-tree imbalance and neighborhood reuse to deliver near hand-tuned speed from V100 through Blackwell-class GPUs. Third, it designs a corresponding set of multi-GPU primitives that extend their single-GPU counterparts to cross-GPU execution, with adaptive task splitting, work stealing, copy-vs-recompute decisions, and topology-aware scheduling, so that a single mining program scales from one GPU to many without rewriting.

Broader Impacts

This project will lower the entry barrier to GPU graph mining for applications in cybersecurity, drug discovery, fraud detection, and AI data pipelines. The PI will release an open-source library of mining primitives, datasets, and a reproducible community benchmark to broaden access to high-performance graph analytics. The project includes a new graduate course on GPU Systems for Data-Intensive Computing, and a summer workshop introducing South Jersey high-school students to data science and GPU computing. The PI will mentor undergraduate and graduate researchers, with explicit recruitment of women and underrepresented students, and engage NVIDIA and AMD graph teams to evaluate new hardware against the released primitives and benchmark.